Experiment 2: Output

Automobile_2a:

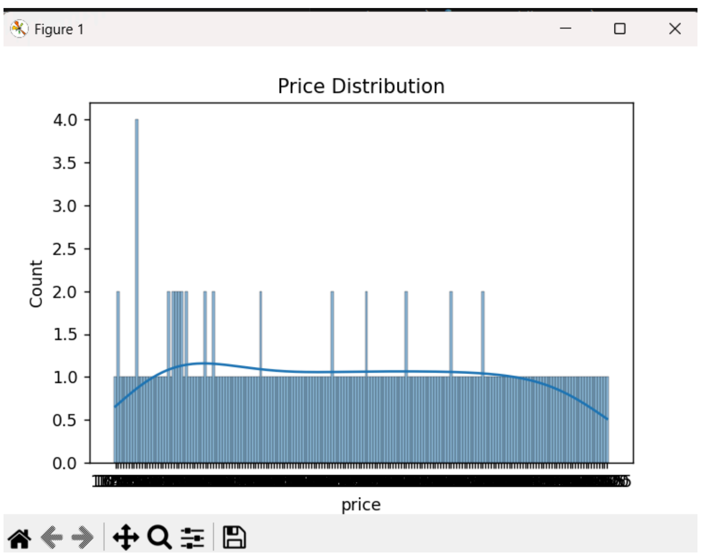
```
  symboling  normalized-losses        make fuel-type  ... peak-rpm city-mpg highway-mpg    price
0         3                NaN  alfa-romero       gas  ...   5000.0       21          27  13495.0
1         3                NaN  alfa-romero       gas  ...   5000.0       21          27  16500.0
2         1                NaN  alfa-romero       gas  ...   5000.0       19          26  16500.0
3         2              164.0         audi       gas  ...   5500.0       24          30  13950.0
4         2              164.0         audi       gas  ...   5500.0       18          22  17450.0

[5 rows x 26 columns]
     symboling  normalized-losses        make fuel-type  ... peak-rpm city-mpg highway-mpg
price
0            3                NaN  alfa-romero       gas  ...   5000.0       21          27  13495.0
1            3                NaN  alfa-romero       gas  ...   5000.0       21          27  16500.0
2            1                NaN  alfa-romero       gas  ...   5000.0       19          26  16500.0
3            2              164.0         audi       gas  ...   5500.0       24          30  13950.0
4            2              164.0         audi       gas  ...   5500.0       18          22  17450.0
..         ...                ...          ...       ... ...      ...      ...         ...      ...
200         -1               95.0        volvo       gas  ...   5400.0       23          28  16845.0
201         -1               95.0        volvo       gas  ...   5300.0       19          25  19045.0
202         -1               95.0        volvo       gas  ...   5500.0       18          23  21485.0
203         -1               95.0        volvo    diesel  ...   4800.0       26          27  22470.0
204         -1               95.0        volvo       gas  ...   5400.0       19          25  22625.0

[205 rows x 26 columns]

Data Types
symboling              int64
normalized-losses    float64
make                     str
fuel-type                str
aspiration               str
num-of-doors             str
body-style               str
drive-wheels             str
engine-location          str
wheel-base           float64
length               float64
width                float64
height               float64
curb-weight            int64
engine-type              str
```

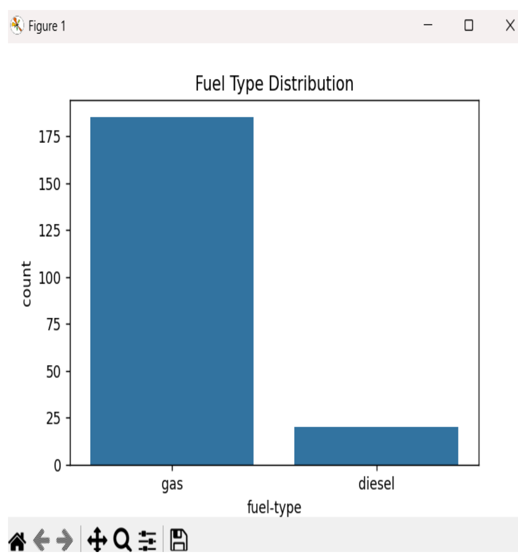
```
num-of-cylinders        str
engine-size           int64
fuel-system             str
bore                float64
stroke              float64
compression-ratio   float64
horsepower          float64
peak-rpm            float64
city-mpg              int64
highway-mpg           int64
price               float64
dtype: object

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 205 entries, 0 to 204
Data columns (total 26 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   symboling          205 non-null    int64
 1   normalized-losses  164 non-null    float64
 2   make               205 non-null    str
 3   fuel-type          205 non-null    str
 4   aspiration         205 non-null    str
 5   num-of-doors       203 non-null    str
 6   body-style         205 non-null    str
 7   drive-wheels       205 non-null    str
 8   engine-location    205 non-null    str
 9   wheel-base         205 non-null    float64
 10  length             205 non-null    float64
 11  width              205 non-null    float64
 12  height             205 non-null    float64
 13  curb-weight        205 non-null    int64
 14  engine-type        205 non-null    str
 15  num-of-cylinders   205 non-null    str
 16  engine-size        205 non-null    int64
 17  fuel-system        205 non-null    str
 18  bore               201 non-null    float64
 19  stroke             201 non-null    float64
 20  compression-ratio  205 non-null    float64
 21  horsepower         203 non-null    float64
 22  peak-rpm           203 non-null    float64
 23  city-mpg           205 non-null    int64
 24  highway-mpg        205 non-null    int64
```

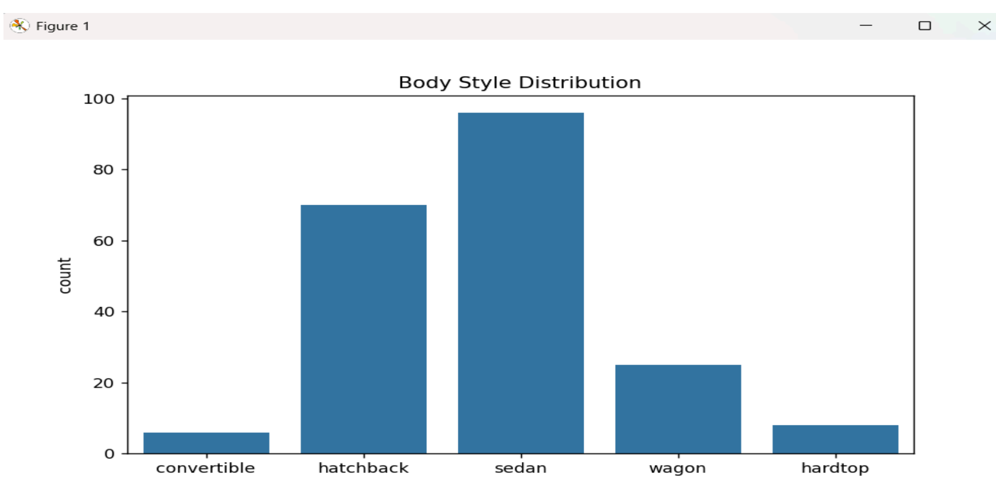
```
 25  price              201 non-null    float64
dtypes: float64(11), int64(5), str(10)
memory usage: 50.2 KB

Shape of Dataset
(205, 26)

Column Names
Index(['symboling', 'normalized-losses', 'make', 'fuel-type', 'aspiration',
       'num-of-doors', 'body-style', 'drive-wheels', 'engine-location',
       'wheel-base', 'length', 'width', 'height', 'curb-weight', 'engine-type',
       'num-of-cylinders', 'engine-size', 'fuel-system', 'bore', 'stroke',
       'compression-ratio', 'horsepower', 'peak-rpm', 'city-mpg',
       'highway-mpg', 'price'],
      dtype='str')

Statistical Summary
        symboling  normalized-losses  wheel-base  ...    city-mpg  highway-mpg         price
count  205.000000         164.000000  205.000000  ...  205.000000   205.000000    201.000000
mean     0.834146         122.000000   98.756585  ...   25.219512    30.751220  13207.129353
std      1.245307          35.442168    6.021776  ...    6.542142     6.886443   7947.066342
min     -2.000000          65.000000   86.600000  ...   13.000000    16.000000   5118.000000
25%      0.000000          94.000000   94.500000  ...   19.000000    25.000000   7775.000000
50%      1.000000         115.000000   97.000000  ...   24.000000    30.000000  10295.000000
75%      2.000000         150.000000  102.400000  ...   30.000000    34.000000  16500.000000
max      3.000000         256.000000  120.900000  ...   49.000000    54.000000  45400.000000

[8 rows x 16 columns]

Missing Values
symboling             0
normalized-losses    41
make                  0
fuel-type             0
aspiration            0
num-of-doors          2
body-style            0
drive-wheels          0
engine-location       0
wheel-base            0
length                0
width                 0
height                0
curb-weight           0
```

engine-type 0
num-of-cylinders 0
engine-size 0
fuel-system 0
bore 4
stroke 4
compression-ratio 0
horsepower 2
peak-rpm 2
city-mpg 0
highway-mpg 0
price 4
dtype: int64

Automobile_2a_part2:







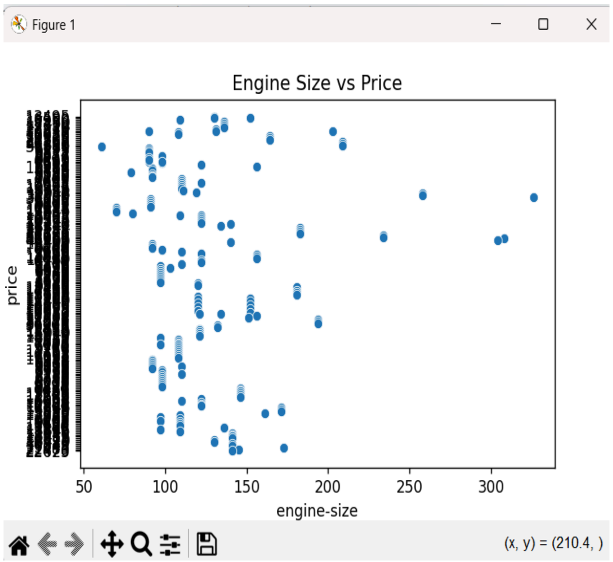


Figure 1
Engine Size vs Price
price
engine-size
50
100
150
200
250
300
(x, y) = (210.4,)

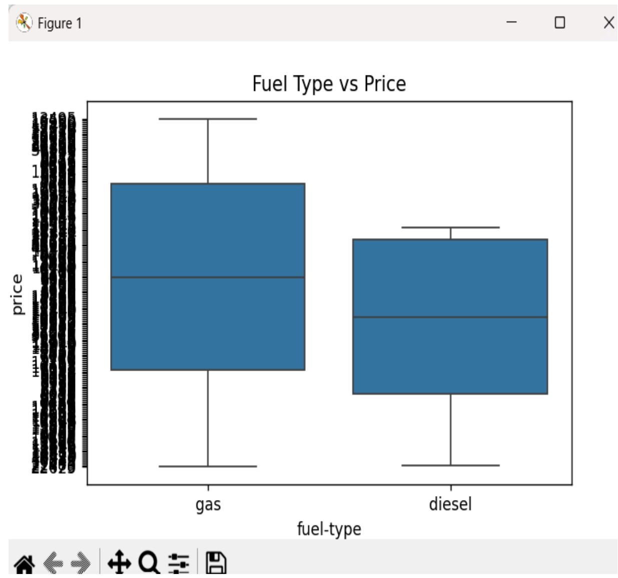
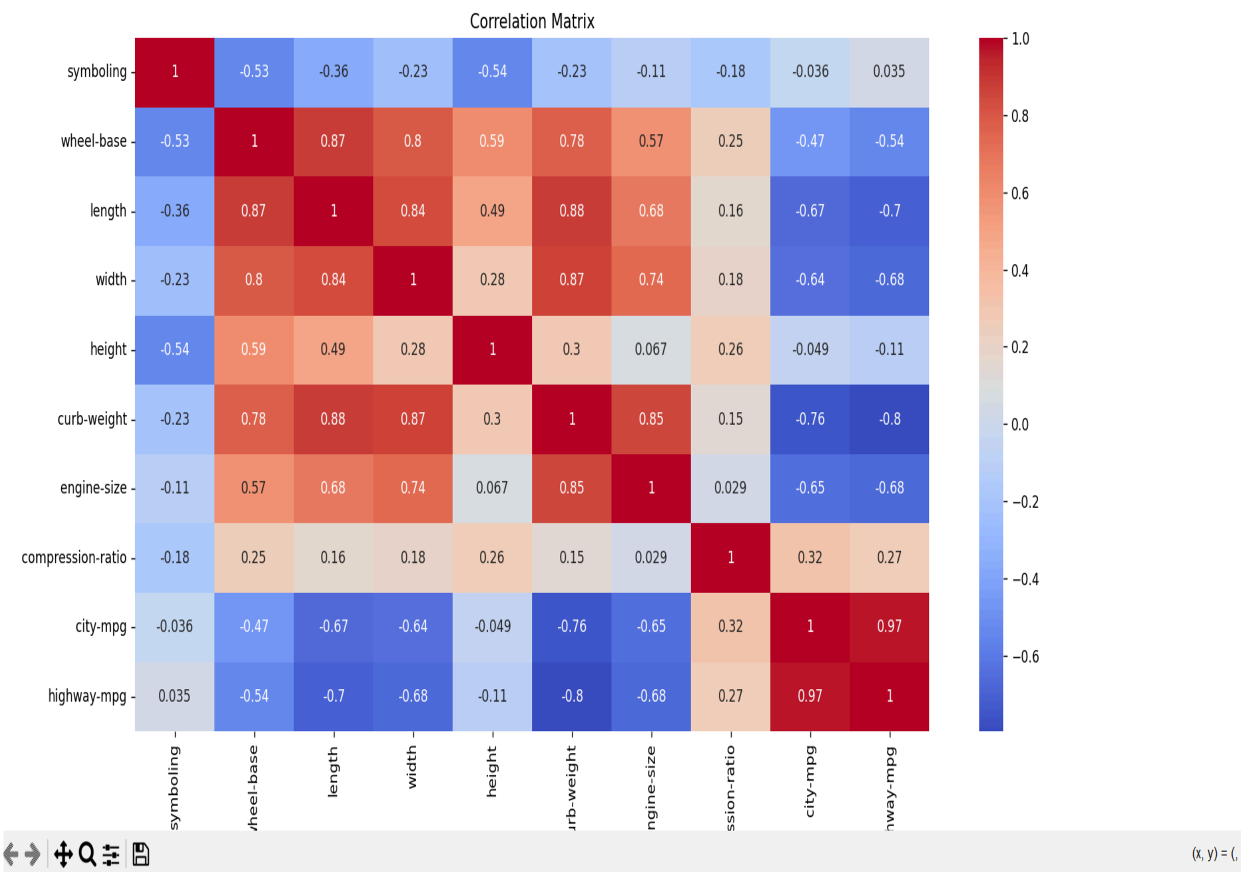


Figure 1
Fuel Type vs Price
price
gas
diesel
fuel-type


Correlation Matrix
symboling 1 -0.53 -0.36 -0.23 -0.54 -0.23 -0.11 -0.18 -0.036 0.035
wheel-base -0.53 1 0.87 0.8 0.59 0.78 0.57 0.25 -0.47 -0.54
length -0.36 0.87 1 0.84 0.49 0.88 0.68 0.16 -0.67 -0.7
width -0.23 0.8 0.84 1 0.28 0.87 0.74 0.18 -0.64 -0.68
height -0.54 0.59 0.49 0.28 1 0.3 0.067 0.26 -0.049 -0.11
curb-weight -0.23 0.78 0.88 0.87 0.3 1 0.85 0.15 -0.76 -0.8
engine-size -0.11 0.57 0.68 0.74 0.067 0.85 1 0.029 -0.65 -0.68
compression-ratio -0.18 0.25 0.16 0.18 0.26 0.15 0.029 1 0.32 0.27
city-mpg -0.036 -0.47 -0.67 -0.64 -0.049 -0.76 -0.65 0.32 1 0.97
highway-mpg 0.035 -0.54 -0.7 -0.68 -0.11 -0.8 -0.68 0.27 0.97 1
symboling
heel-base
length
width
height
urb-weight
ngine-size
ssion-ratio
city-mpg
hway-mpg
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
(x, y) = (,)

exp-2_Automobile_2b:

Numeric Attributes
```
Index(['symboling', 'wheel-base', 'length', 'width', 'height', 'curb-weight',
       'engine-size', 'compression-ratio', 'city-mpg', 'highway-mpg'],
      dtype='str')
```

Nominal Attributes
['make', 'fuel-type', 'aspiration', 'body-style', 'drive-wheels', 'engine-location', 'engine-type', 'num-of-cylinders', 'fuel-system']

Binary Attributes
['fuel-type', 'aspiration', 'engine-location']

exp-2_Automobile_2c:

Nominal Attribute Dissimilarity
Fuel Type vs Aspiration = 1.0

exp-2_Automoblie_2d:
Euclidean Distance
3005.0

Euclidean Dissimilarity
-29.05

Manhattan Distance
3005.0

manhattan Dissimilarity
-29.05

Minkowski Distance
3004.9999999999986

minkowski Dissimilarity
-29.049999999999986

Cosine Similarity
0.9999984612606099

Cosine Dissimilarity
1.5387393901278301e-06